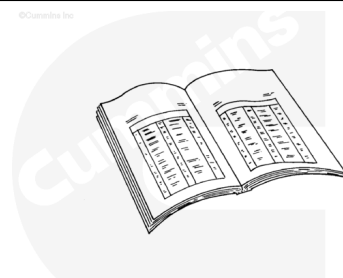


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

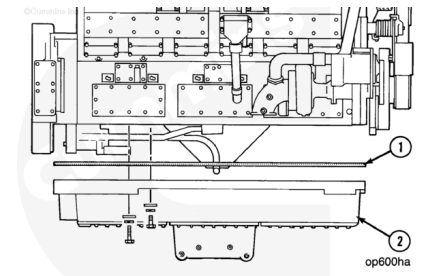
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Remove the fuel filter, if needed. Refer to Procedure 006-015 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html>)
- Remove the brackets for the marine gear oil cooler plumbing, if equipped. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)
- Remove the oil pan from the oil pan adapter.

- Remove the cover plates from the oil pan adapter, if needed.

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the capscrews, lock washers, and plain washers.

Remove the oil pan adapter assembly (2) and gasket (1).

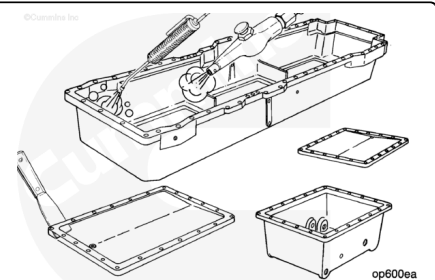
Discard the gasket.

Note : The clearance must be sufficient from the oil suction tube before the adaptor can be pulled from the bottom of the engine.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

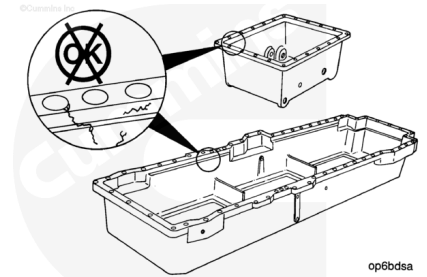
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a scraper. Clean all gasket surfaces.

Use steam. Clean the parts. Dry with compressed air.

Make sure all dirt and debris is cleaned from the magnets on the drain plugs.

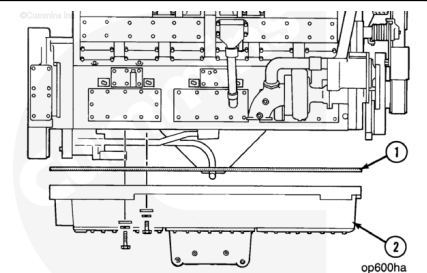
Check the parts for cracks or other damage. If any of the parts are cracked or otherwise damaged, they **must** be replaced



Install

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ CAUTION ⚠

Do not damage the oil suction tube when installing the adapter assembly. If the part is damaged, a major engine failure can result from the lack of lubrication.

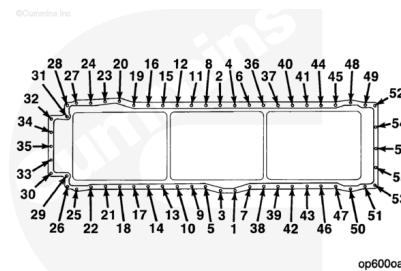
Note : The gaskets are manufactured from a material that enlarges when it contacts engine oil. Do not use gasket shellac or cement to install the gaskets, as a leak will result. If necessary, use a light layer of grease to secure the gaskets.

Install the gasket (1) on the adapter (2). Install the adapter to the cylinder block with the plain washers, lock washers, and capscrews.

Note : The torque sequence shows a K38 oil pan adapter. The K50 pan adapter contains more capscrews. Use the same method to tighten the capscrews on the K50 oil pan adapter.

Tighten the capscrews in the sequence shown in the illustration.

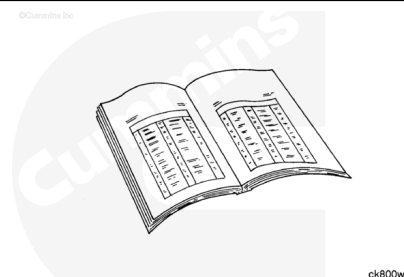
Torque Value: 60 N·m [44 ft·lb]



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the oil pan onto the oil pan adapter.
- Install the cover plates on the oil pan adapter, if removed.
- Install the brackets for the marine gear oil cooler plumbing, if equipped. Refer to Procedure 008-137 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-137-tr.html>)
- Install the new fuel filters, if removed. Refer to Procedure 006-015 in Section 6. (</qs3/pubsys2/xml/en/procedures/28/28-006-015-tr.html>)
- Install the new lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-037-tr.html>)
- Operate the engine at idle speed and inspect for oil leaks.

